Assignment 3: Understanding Customer Tipping Behavior

Dataset: **Tips** (seaborn). In this notebook we clean, explore, and visualize the data to answer management's questions about tipping behavior.

Q1: Loading and Understanding the Dataset (10 marks)

In [2]:
```
# import lib
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

sns.set_style("whitegrid")
%matplotlib inline
```

In [4]:
```
# Q1.1: Load dataset
tips = sns.load_dataset('tips')
```

In [5]:
```
# Q1.2: Display first rows
tips.head()
```

Out[5]:

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

In [6]:
```
# Q1.3: Dataset shape and columns
print("Dataset shape:", tips.shape)
print("Column names:", list(tips.columns))
```

```
Dataset shape: (244, 7)
Column names: ['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size']
```

Q2: Basic Filtering – Who Tips More? (25 marks)

In [7]:
```
# Q2.1: Weekend
weekend = tips[tips['day'].isin(['Sat', 'Sun'])]

# Weekday
weekdays = tips[tips['day'].isin(['Thur', 'Fri'])]
```

```
print("Weekend rows:", weekend.shape[0])
print("Weekday rows:", weekdays.shape[0])
```

```
Weekend rows: 163
Weekday rows: 81
```

In [8]:
```
# Q2.2: Average tip, Average total bill
weekend_avg_tip = weekend['tip'].mean()
weekend_avg_bill = weekend['total_bill'].mean()

weekday_avg_tip = weekdays['tip'].mean()
weekday_avg_bill = weekdays['total_bill'].mean()

print(f"Weekend  -> Avg tip: {weekend_avg_tip:.2f}, Avg total bill: {weekend_avg
print(f"Weekday  -> Avg tip: {weekday_avg_tip:.2f}, Avg total bill: {weekday_avg
```

```
Weekend  -> Avg tip: 3.12, Avg total bill: 20.89
Weekday  -> Avg tip: 2.76, Avg total bill: 17.56
```

Answer Q2.3: Compare this with weekday customers.

Weekend customers (Saturday & Sunday) leave a higher average tip (≈ $3.12) than weekday customers (≈ $2.76). Weekend total bills are also higher on average (≈ $20.89 vs ≈ $17.56), which partly explains the higher tips, since tip amount tends to scale with the bill. This suggests weekends bring larger groups or pricier orders (e.g., dinner rather than lunch), driving both higher spending and higher tips.

Q3: Sorting and Identifying High-Value Customers (15 marks)

In [10]:
```
# Q3.1: Sort customers by total_bill (descending)
sorted_tips = tips.sort_values('total_bill', ascending=False)
```

In [11]:
```
# Q3.2: Display the top 10 highest bills
top10 = sorted_tips.head(10)
top10
```

Out[11]:

	total_bill	tip	sex	smoker	day	time	size
170	50.81	10.00	Male	Yes	Sat	Dinner	3
212	48.33	9.00	Male	No	Sat	Dinner	4
59	48.27	6.73	Male	No	Sat	Dinner	4
156	48.17	5.00	Male	No	Sun	Dinner	6
182	45.35	3.50	Male	Yes	Sun	Dinner	3
102	44.30	2.50	Female	Yes	Sat	Dinner	3
197	43.11	5.00	Female	Yes	Thur	Lunch	4
142	41.19	5.00	Male	No	Thur	Lunch	5
184	40.55	3.00	Male	Yes	Sun	Dinner	2
95	40.17	4.73	Male	Yes	Fri	Dinner	4

Answer Q3.3: What do you notice about their group sizes?

The top 10 highest-bill groups have an average size of about 3.8 people, noticeably larger than the overall average group size of about 2.57. This makes sense: bigger groups order more food and drinks, which naturally pushes the total bill higher. So high-value bills are strongly associated with larger party sizes rather than being random outliers.

Q4: Data Quality Check & Cleaning (10 marks)

In [12]:
```
# Q4.1: missing values
print(tips.isnull().sum())
print("\nTotal missing values:", tips.isnull().sum().sum())
```

```
total_bill    0
tip           0
sex           0
smoker        0
day           0
time          0
size          0
dtype: int64

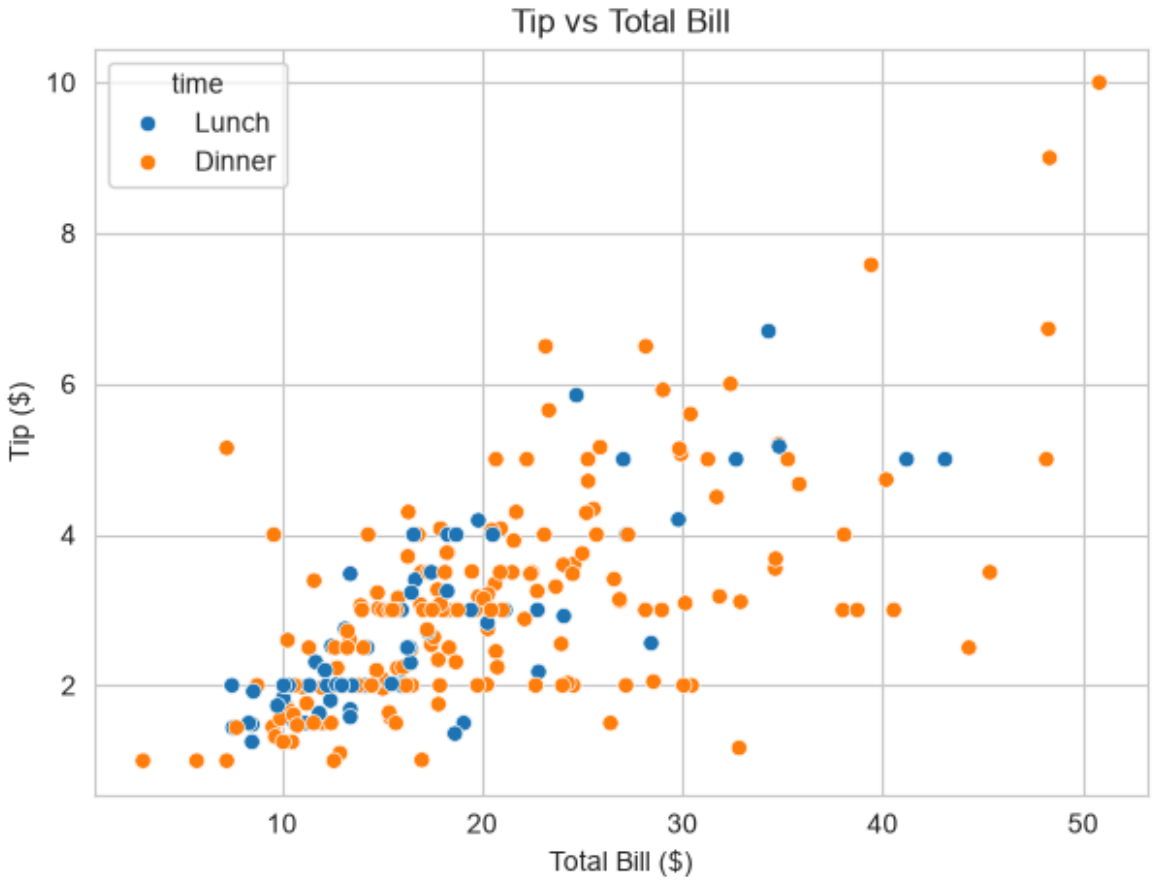
Total missing values: 0
```

Answer Q4.2: Why missing values are dangerous in machine learning.

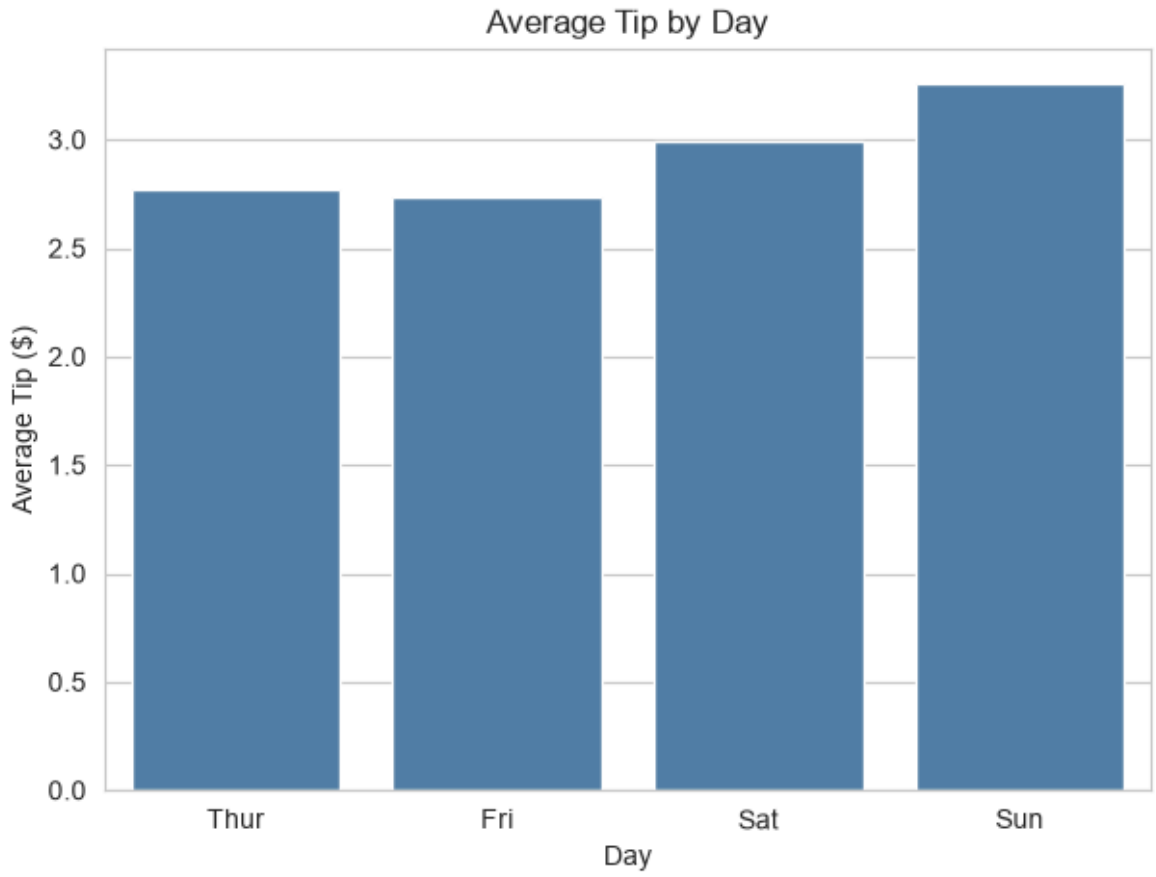
Missing values are dangerous because most machine learning algorithms cannot process NaN values directly and will throw errors or produce unreliable results. If missing data is simply dropped, it can shrink the dataset and introduce bias if the missingness is not random (e.g., certain groups systematically not reporting tips). If missing values are filled incorrectly, they can distort statistics like the mean or variance, mislead the model into learning false patterns, and reduce the model's predictive accuracy. Proper handling (imputation, removal, or flagging) is essential to keep the analysis and any downstream model trustworthy. In this dataset specifically, there are no missing values, so no cleaning is required here.

Q5: Visualization – What Influences Tips? (25 marks)

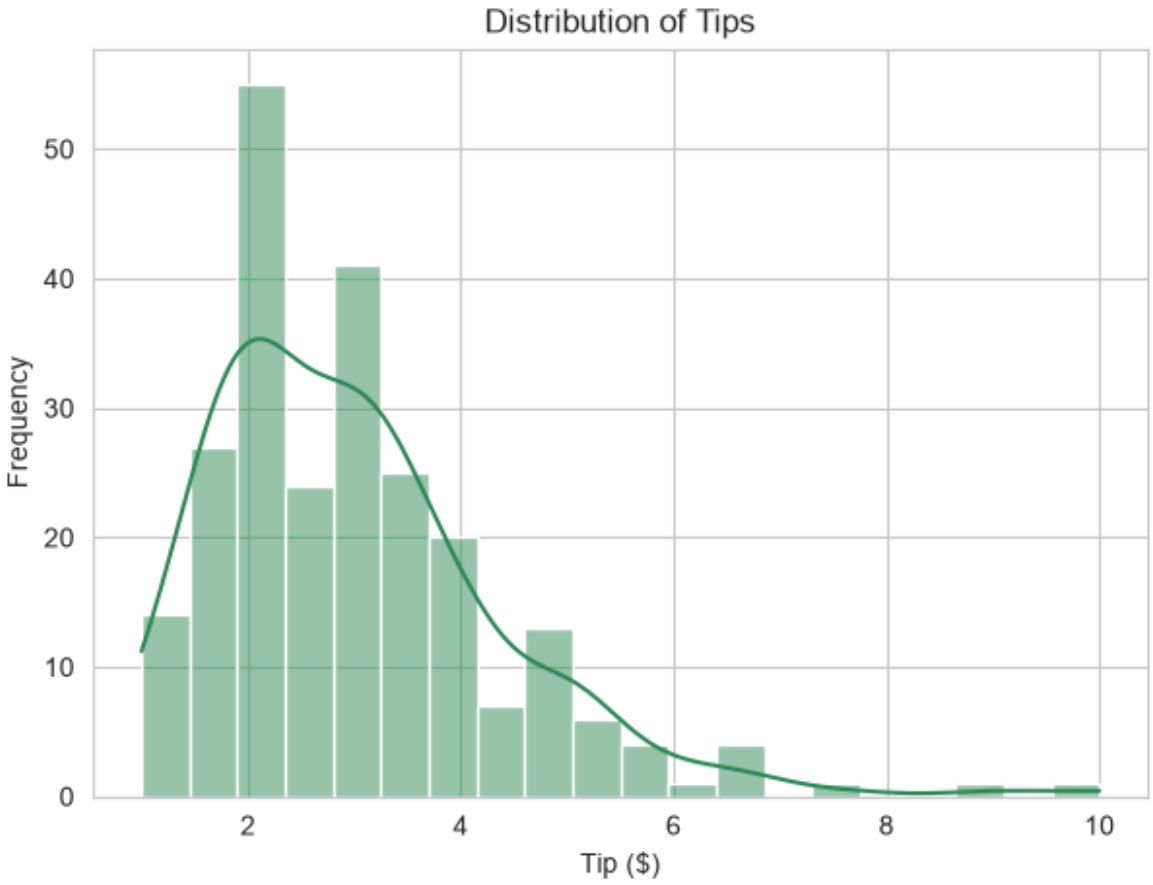
In [13]:
```
# Q5.1: Tip vs Total Bill (scatter plot)
plt.figure(figsize=(7,5))
sns.scatterplot(data=tips, x='total_bill', y='tip', hue='time')
plt.title('Tip vs Total Bill')
plt.xlabel('Total Bill ($)')
plt.ylabel('Tip ($)')
plt.show()
```



In [14]:
```
# Q5.1: Average tip by day (bar plot)
plt.figure(figsize=(7,5))
avg_tip_by_day = tips.groupby('day', observed=True)['tip'].mean().reindex(['Thur
sns.barplot(x=avg_tip_by_day.index, y=avg_tip_by_day.values, color='steelblue')
plt.title('Average Tip by Day')
plt.xlabel('Day')
plt.ylabel('Average Tip ($)')
plt.show()
```



In [15]:
```
# Q5.1: Tip distribution (histogram)
plt.figure(figsize=(7,5))
sns.histplot(tips['tip'], bins=20, kde=True, color='seagreen')
plt.title('Distribution of Tips')
plt.xlabel('Tip ($)')
plt.ylabel('Frequency')
plt.show()
```



Answer Q5.2: Interpret each plot in 1–2 sentences.

- **Tip vs Total Bill (scatter plot):** There is a clear positive relationship between total bill and tip (correlation ≈ 0.68) — as the bill increases, the tip tends to increase too, which is expected since tips are usually a percentage of the bill.
- **Average tip by day (bar plot):** Sunday has the highest average tip (≈ $3.26), followed by Saturday, while Thursday and Friday have the lowest average tips, supporting the idea that weekend dining leads to higher tips.
- **Tip distribution (histogram):** Tip amounts are right-skewed, with most tips clustering between $2 and $4, and a long tail of a few larger tips (up to about $10), indicating that very generous tips are relatively rare.

Q6: Final Insight & Conclusion (25 marks)

Answer Q6:

Customers — Male customers leave a slightly higher average tip in dollars ($3.09 vs $2.83 for females), but when measured as a percentage of the bill, women actually tip a bit more generously (≈16.6% vs ≈15.8%). Smoking status makes almost no difference to tipping behavior (≈16.3% for smokers vs ≈15.9% for non-smokers) — too small a gap to be a meaningful driver. The strongest factor by far is the total bill: tip amount rises steadily as the bill increases (correlation ≈ 0.68), and larger groups tend to generate both bigger bills and bigger tips.

Restaurant managers — Tips are highest on weekends, especially Sunday, both in dollar amount and reflecting bigger group sizes and higher spending. Gender and smoking status are minor factors and not reliable levers for predicting or influencing tip size. Since tip size tracks bill size so closely, management's best strategy for increasing total tip revenue is to focus on strategies that increase the total bill (upselling, larger portions, combo deals) and to ensure adequate staffing on Saturdays and Sundays, when both traffic and tipping are highest. This insight can also help with staff scheduling, shift incentives, and predicting server earnings across the week.

In [ ]: